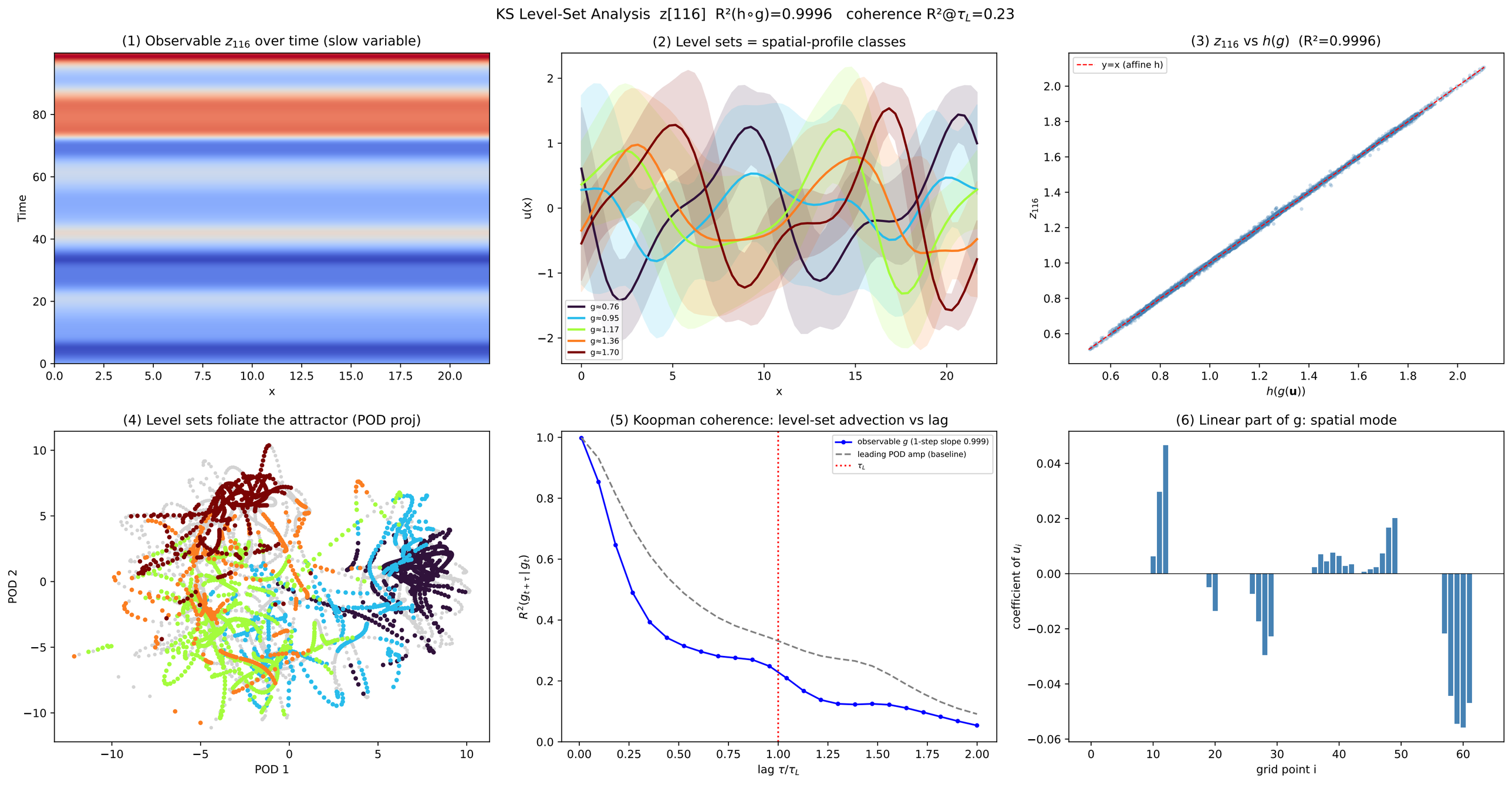

KS Level-Set Analysis  z[116]  R²(h∘g)=0.9996  coherence R²@τ_L=0.23
(1) Observable z_116 over time (slow variable)
Time
x
(2) Level sets = spatial-profile classes
u(x)
x
g≈0.76
g≈0.95
g≈1.17
g≈1.36
g≈1.70
(3) z_116 vs h(g)  (R²=0.9996)
y=x (affine h)
z_116
h(g(u))
(4) Level sets foliate the attractor (POD proj)
POD 2
POD 1
(5) Koopman coherence: level-set advection vs lag
observable g (1-step slope 0.999)
leading POD amp (baseline)
τ_L
R²(g_{t+τ} | g_t)
lag τ/τ_L
(6) Linear part of g: spatial mode
coefficient of u_i
grid point i